Lesson 1: Charge

ChatGPT edition — make an invisible interaction repeatable

Conservation, sign, and polarization

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

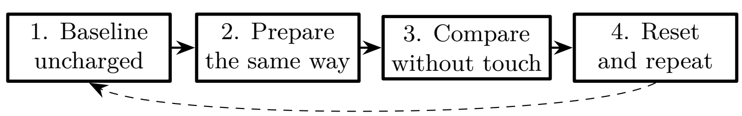
- Transparent tape and a smooth table
- Two pencils and two cups as supports
- Balloon or plastic comb; wool cloth
- Empty, dry aluminum can
- Ruler, scissors, and notebook
- Optional: hygrometer

Predict first

Before touching the materials, choose: two strips prepared alike will ◯ attract ◯ repel ◯ do nothing. A charged comb will move a neutral can because ______________________________

The evidence plan

Static charge is easily lost and easy to imagine. This investigation therefore uses a control, repeats each comparison, and changes one thing at a time.

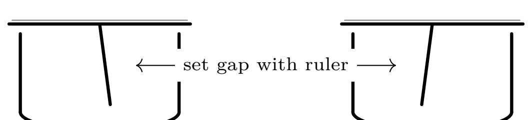


Questions that keep the test honest

What is held constant: strip length, preparation, suspension, and separation. What changes: only the pair being compared. What counts as evidence: direction of first motion, not where the strips finish after touching.

Part A — establish a baseline

1. Cut four tape strips 15 cm long. Fold a 2 cm handle on each. Label two strips S_1, S_2; leave two for the stacked test.
2. Make two suspension stands: bridge a pencil across a cup and hang a strip from its handle. Put the stands 10 cm apart.
3. Suspend S_1 and S_2 without rubbing or peeling them. Bring the stands together until the strips are 2 cm apart. Do they move before air currents settle? ◯ yes ◯ no
4. Hold a hand 10 cm away, then withdraw it. If the strips swing, wait until they stop. This is the motion floor for later comparisons.



Baseline: supports isolate the tape from hands and table

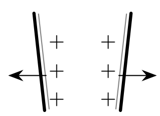
Baseline checkpoint. Continue only when the unprepared strips show no repeatable first motion across two trials. Otherwise discharge both, reduce drafts, and repeat.

Part B — same preparation, then opposite histories

B1. Two strips prepared alike

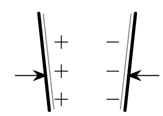
1. Press S_1 flat on the same clean table area ten times with a finger. Pull it up briskly by its handle. Touch only the folded handle.
2. Repeat exactly with S_2. Suspend both immediately.
3. Starting at 8 cm, reduce the gap by 1 cm every three seconds. At what gap does the first repeatable motion appear? Record its direction.
4. Discharge, prepare again, and perform three trials. Reverse the left/right stands on trial 3. Does the direction follow the strips?

State 1: like histories



repulsion identifies like charge signs

State 2: unlike histories

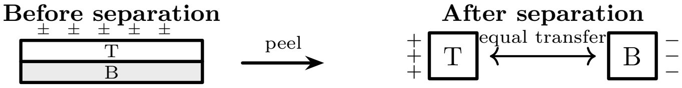


attraction is consistent with opposite signs, but not proof alone

B2. Separate a stacked pair

5. Put one fresh strip on the table and label its handle B (bottom). Press a second strip directly on top and label it T (top).
6. Lift the pair together by B. Pause: before separation, predict the sign relationship between T and B: ____________
7. Pull T and B apart in one smooth motion. Suspend them without letting them touch your clothes, hair, or each other.
8. Compare T with T from a repeated pair, B with B, then T with B. Reset between comparisons.

Before separation **After separation**



sign labels are conventional; the opposite behavior is observable

Intermediate questions
Why is attraction alone ambiguous? Could a charged strip attract an uncharged one? Which result is stronger evidence for like signs: attraction or repulsion? If T gains charge, where must an equal amount of opposite charge remain?

Record the tape evidence

Comparison	Trial	First motion	Threshold gap	Notes: touch, draft, delay
S_1 vs S_2	1			
same preparation	2			
	3 reversed			
T vs T	1			
B vs B	1			
T vs B	1			

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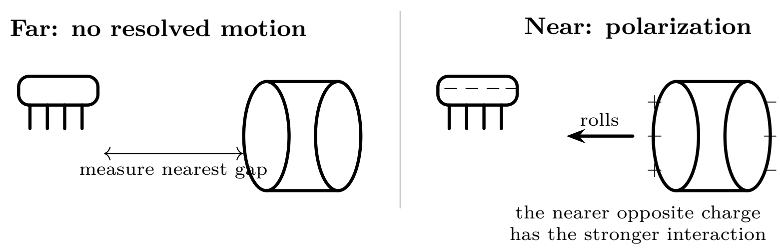
Part C — a charged object and a neutral can

1. Lay the empty can on a level table. Mark its starting line. Confirm that it remains still for ten seconds with the comb 30 cm away.
2. Rub the comb with wool exactly ten strokes. Keep the same hand, stroke length, and comb face for every trial.
3. Hold the comb parallel to the can, 10 cm from its nearest surface. Do not touch. Move inward in 1 cm steps, pausing three seconds.
4. Record the first distance at which the can begins rolling toward the comb. Stop the comb; does the can continue? Move it sideways; does the can track it?
5. Return the can to its line and repeat three times. Recharge the comb with ten strokes before every trial.
6. Control: repeat the distance sequence after touching the comb to your hand and table. A smaller or absent response supports the charge claim.



Trial	Charged threshold	Discharged control	Tracks sideways?
1	cm	cm / none	
2	cm	cm / none	
3	cm	cm / none	

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ChatGPT edition — Lesson 1 verification

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Troubleshoot without changing the story

Observation	Likely cause	One controlled response
Nothing moves	Humidity, charge leakage, gap too large	Dry surfaces; shorten delay; record humidity; prepare again identically.
Both strips drift together	Draft or moving supports	Close windows; wait; swap stands left-to-right.
Strips first attract, then cling	They touched and charge transferred	Use only the first motion; discharge and reset.
Can rolls before approach	Sloped or vibrating table	Rotate can, level the surface, repeat baseline.
Only one trial works	Preparation or timing changed	Count strokes and use a three-second cadence.

Build the proof

1. **Repulsion test:** Two independently prepared alike strips repelled in ______ of three trials. Repulsion rules out a merely neutral partner and is evidence that the strips carried like signs.
2. **Opposite-history test:** T-T and B-B showed ____________; T-B showed ____________. The consistent pattern supports two charge signs.
3. **Conservation claim:** Separating the joined system redistributes charge. The evidence does *not* show charge appearing from nothing: opposite changes occur on the two members of the original pair.
4. **Polarization test:** The charged comb moved the neutral can from a median threshold of ______ cm, while the discharged control ______________. Because the can began neutral, attraction alone does not establish opposite net charge; charge shifted within it.

Verification questions

1. Which single observation best distinguishes like charge from polarization? Why?
2. What result followed the object when the supports were swapped?
3. Does a smaller threshold distance mean a stronger or weaker observed interaction under this procedure?
4. What uncontrolled variable most limits your claim?
5. Write one sentence that separates observation from explanation: "I observed ____________________________; this supports ____________________________ because ________________________."

Explain from the evidence

Like charge signs repel and unlike signs attract. Peeling and rubbing transfer charge between materials; for the combined system, charge is conserved. A charged object can attract a neutral conductor because mobile charge in the conductor redistributes. The nearer opposite-sign region experiences the stronger interaction. Humidity and contaminated surfaces provide leakage paths, so null results belong in the record.

Change one thing

Repeat the can trial at two measured humidity levels, or compare two insulating materials. Keep geometry, stroke count, timing, and observer fixed. Plot the three threshold distances rather than reporting only the strongest trial.

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.